

There is no doubt that India has successfully rocketed itself as a fore-runner in the global automobile industry, and there is also no questioning the fact that the country favours two-wheelers. So far, internal combustion engine (ICE) and ICE two-wheeler have been the country's favourable mode of mobility and its electric counterpart has not yet made much of a dent in the market.

Electric two-wheelers represented only one percent of the total two-wheeler sales in the country in 2020, which stood at 150,000 units. The difference is jarring, which begs the question: What do electric players need to do to get the same love from the market?

Starts with the customer

A look at the market today presents us with a mixed picture. While the EV industry is picking up pace in India, it is still taking baby steps compared to the pace at which the other global markets are moving. A quick look back and we realise that ICE two-wheeler owners were and still are presented with very few options, whereas electric alternatives come with a wide range of variety.

Scooters and motorcycles are the two sub-segments within the two-wheeler segment that is most popular in India and also in most other parts of the world. While scooters constitute 29 percent of all two-wheeler production in India, motorcycles make up 67 percent.

Yet most electric two-wheeler players have set their eyes upon the scooter segment, leaving the motorcycle market inadequately touched. This means that 67 percent of potential two-wheeler buyers do not have a viable alternative in the electric segment, at least as of now. This, in itself, is a major setback for the average customer that wishes to have a motorcycle but has to reluctantly settle for a scooter.

Given its size and market, India is undeniably a potential hub for electric two-wheelers, but a common and somewhat frustrating hindrance that almost every EV player faces in the country is customer hesitancy. Knowing what the potential two-wheeler buyer is looking at when evaluating an

Some common questions buyers want affirmative answers to while choosing an electric two-wheeler (E2W):

- Is the E2W comparable to an ICE two-wheeler in terms of capability/product features?
- Is an E2W at price parity with ICE vehicles? Will financing be easily available for it?
- Is adequate and accessible charging infrastructure available?
- Does E2W have strong enough battery to meet daily commute requirements?
- Is after-sale service available? What about resale value? Any technology risk?

electric alternative is key to a successful transition in the market.

Customers are also on the lookout for the model with the most advanced features. With the market now filling with IoT-enabled and AI-enabled two-wheelers, it can be said that more feature-rich variants are going to be the norm for customers to opt for an electric two-wheeler.

"The newer generation is moving towards how much they can do and how many features are there in terms of the technology platforms that make navigation and access much easier," says Mohd Sahil Ali, Senior Sustainability Analyst, IHS Markit.

Financing is a major headwind in the adoption of electric two-wheelers. With an already exorbitant upfront price, consumers are also riddled with the uncertainty of the after-sale service and resale value of the vehicle after half a decade or so.

The consumers also need some kind of reassurance towards after-sale service and resale value of vehicle after four to five years. Residual value is also a key concern for the financiers, especially in the backdrop of declining battery pack prices and the seven-year to eight-year battery life, which could significantly impact the residual value of repossessed vehicles.

All these factors combine to further the hesitancy among consumers to switch over to electric mobility. Samsher Dewan, Vice President & Group Head—Corporate Sector Ratings, ICRA, suggests, "Industry participants must come out with innovative solutions which could be some kind of a buyback scheme or some leasing options which could help us in increasing the financing penetration."

While financing is required for 45 to 50 percent for the overall two-wheeler segment, the same stands at merely 10 percent for the electric

two-wheeler segment. A key reason for that, experts believe, is their lack of confidence in the electric segment.

The only way to break this hesitancy is to educate not only customers but also such financiers and taking them into confidence. One way to do that collectively, says Dewan, is to share data on the actual usage of these vehicles with the financiers to give them an idea as to how this segment is actually panning out.

Another weighing factor for customers is the load carrying capability. As per ICRA's research, 55 percent of electric two-wheelers in the market today have lower load capacity, exceeding which may have some implications on the performance of the vehicle, especially with two or more riders.

The price discrepancies between an ICE model and an electric one is perhaps the biggest factor that prevents customers from making the switch.

There is much debate on the total cost of ownership of an electric two-wheeler and its benefits in the long term. However, it is important to realise that a two-wheeler is the vehicle of choice for usually low to middle-income families, for whom initial price is an important factor. They are unlikely to empty their wallets for a product priced substantially higher than the petrol-run alternatives, no matter how great the long-term benefits.

Sensing this big hurdle, the government of India launched the Faster Adoption and Manufacturing of Hybrid and Electric Vehicles in India (FAME India) scheme with an aim to promote the adoption of EVs in the country. Its recent amendments in Phase-II of the policy has garnered much praise from the industry.

But incentives and subsidies, although an encouraging head starter, are not viable forever. Reducing the